Department:
Data Analytics and Mathematical Sciences

Mini Project report on:

"Indigo Sales Analysis For The Next Six Months"

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INTRODUCTION

This project focuses on analyzing data using Microsoft Excel to derive meaningful insights.
The purpose of the project is to understand how Excel functions, formulas, and data visualization tools can be used to interpret real-world datasets.

The motivation behind the project is to gain hands-on experience in:

- Data cleaning
- Applying analytical Excel functions
- Creating charts and visualizations
- Drawing insights from trends and patterns
- Airline revenue modeling
- Cost estimation
- Excel-based forecasting
- Data visualization and interpretation

This project helps build practical skills that are essential for academic, business, and analytical work.

OBJECTIVES

- To collect and organize route-level financial data including aircraft type, flights per year, revenue, and cost.
- Predicting growth of sales through new long haul international flights
- Apply Excel formulas and functions to process and analyze the dataset
- Visualize data using charts and graphs for better understanding.
- Interpret insights and present meaningful conclusions from the findings.

DATASET DESCRIPTION

The dataset was created manually using airline operational assumptions such as:

- Aircraft seat configuration
- Estimated load factor
- Annual flights
- Fare assumptions
- Operational cost estimates

Column name	**Description**	**Data type**
Route	Origin - Destination Project	text
Airline Type	Aircraft model	text
Flights/Year	Number of flights annually	number
Annual Revenue	Estimated revenue from all seats	number
Economy Revenue	Revenue from economy seats only	number
Stretch Revenue	Revenue from premium/business seats	number
Annual Cost	Estimating operating cost of that route	number
Profit/Loss	Revenue- cost	number
Margin	Profit percentage	percentage

METHODOLOGY/EXCEL FEATURES

- The methodology for this project involved a structured process of data collection, preparation, analysis, and forecasting using Microsoft Excel. The first step was to gather accurate and relevant financial information from IndiGo's **Annual Reports for FY 2022–23, FY 2023–24, and FY 2024–25**. These reports were essential sources because they include consolidated and standalone financial statements, operational metrics, passenger numbers, fleet size, and revenue breakups. Specifically, the **Standalone Financial Statements** provided clear insights into IndiGo's operational revenue, expenses, fuel costs, fleet-related costs, and other key financial indicators required for revenue modelling.
- Along with the financial data, the project also incorporated details about **newly launched international and long-haul routes**. These routes were collected from the annual reports, investor presentations, and official route announcements included within these documents. For each new route, information such as aircraft type, seat configuration, flight frequency, and route category (short, medium, long haul) was extracted. This dataset helped us understand how these newly added routes could contribute to revenue generation over the next six months.
- Once all data was collected, it was organized into an Excel spreadsheet with key fields such as Route, Aircraft Type, Flights per Year, Annual Revenue, Economy Revenue, Stretch/Business Revenue, Annual Cost, Profit/Loss, and Margin. The next phase involved calculating estimated revenues using seat capacity, load factor assumptions, average fares, and aircraft-specific premium seat pricing. Costs were also estimated based on aircraft type, flight duration category, and industry-standard operating cost approximations.
- To analyze the data, Excel formulas such as **SUM**, **AVERAGE**, **IF**, **COUNTIF**, and percentage calculations were used extensively to compute revenue, cost, and profitability.n existing annual revenue patterns. Excel features such as **conditional formatting** were used to highlight profitable and loss-making routes, **sorting and filtering** helped categorize flights by profitability and aircraft type.
- For visualization, charts such as bar graphs, line charts, and route comparison visuals were created to help interpret trends and present the findings clearly. Pivot tables were also used to summarize results by aircraft type and route category. Overall, the methodology combined

financial analysis, forecasting, and visualization to evaluate how long-haul and new international routes are likely to impact IndiGo’s revenue soon.

DATA ANALYSIS AND VISUALIZATION

Data Cleaning -

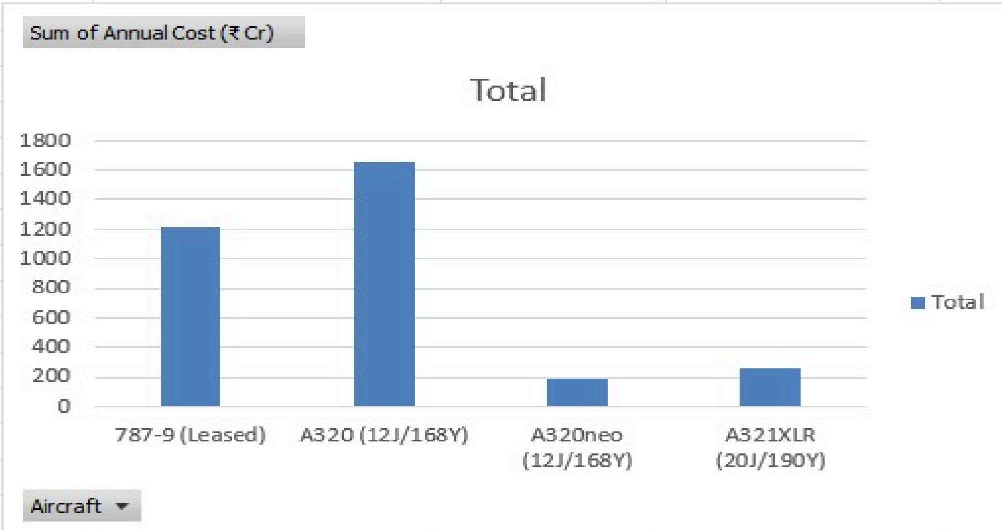
- Verified aircraft configurations
- Standardized all revenue and cost into crores (₹ Cr)
- Ensured numerical fields were formatted as numbers
- Removed text formatting errors

Calculations Performed -

- Annual revenue calculated from estimated load factors and fares
- Annual cost estimated based on aircraft type and distance category
- Profit/Loss derived using arithmetic formulas
- Margin calculated as a percentage
- Revenue and number of flights in a year to a specific route and destination

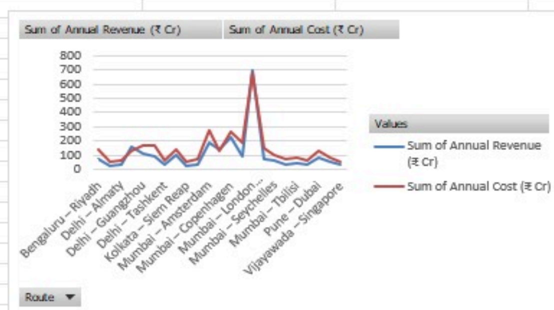
Visualisations -

Row Labels	Sum of Annual Cost (₹ Cr)
787-9 (Leased)	1211.91
A320 (12J/168Y)	1651.31
A320neo (12J/168Y)	185.57
A321XLR (20J/190Y)	255.15
Grand Total	**3303.94**



- Annual operating cost of the various aircrafts

Row Labels	Sum of Annual Revenue (₹ Cr)	Sum of Annual Cost (₹ Cr)
Bengaluru – Riyadh	74.32	135.01
Bhubaneswar – Abu Dhabi	28.93	57.3
Delhi – Almaty	36.61	65.94
Delhi – Athens	157.98	129.28
Delhi – Guangzhou	106.82	167.11
Delhi – Hanoi	88.8	172.67
Delhi – Tashkent	34.88	62.85
Kolkata – Guangzhou	96.75	136.51
Kolkata – Siem Reap	23.96	49.06
Mumbai – Almaty	33.21	70.68
Mumbai – Amsterdam	187.2	271.36
Mumbai – Athens	143.63	125.87
Mumbai – Copenhagen	222.08	265.67
Mumbai – Denpasar (Bali)	89.96	190.74
Mumbai – London (Heathro	697.59	674.88
Mumbai – Madinah	68.42	152.53
Mumbai – Seychelles	61.22	97.36
Mumbai – Tashkent	32.44	68.94
Mumbai – Tbilisi	41.02	79.49
Pune – Bangkok	35.13	63.57
Pune – Dubai	77.97	131.29
Thiruvananthapuram – Mal	57	79.49
Vijayawada – Singapore	29.44	56.34
SUM	**2425.36**	**3303.94**



- Line chart to visualize the difference between the annual revenue and cost

Row Labels	Sum of Annual Revenue (₹ Cr)	Sum of Annual Cost (₹ Cr)
Bengaluru – Riyadh	74.32	135.01
Bhubaneswar – Abu Dhabi	28.93	57.3
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Delhi – Athens	157.98	129.28
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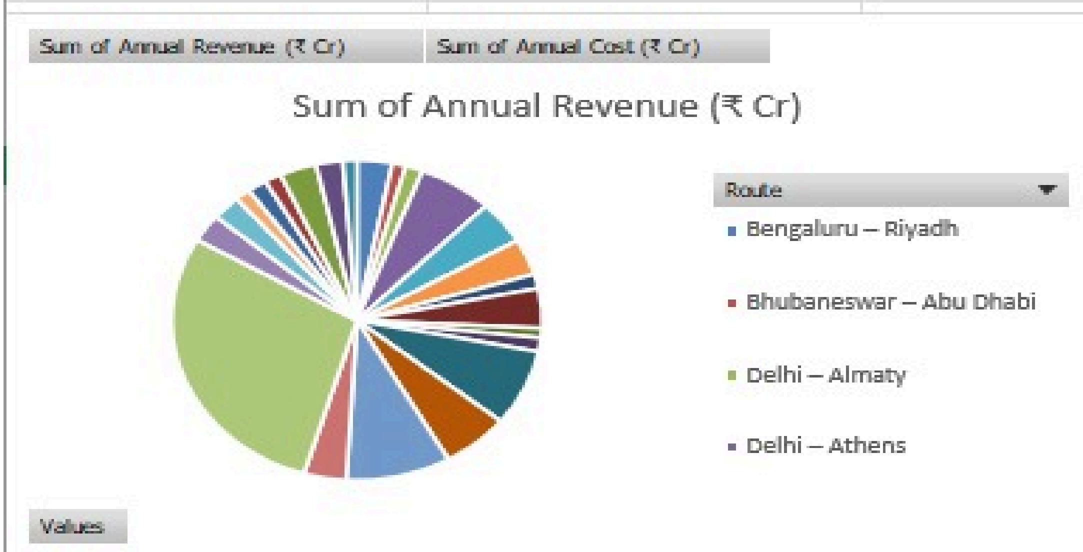
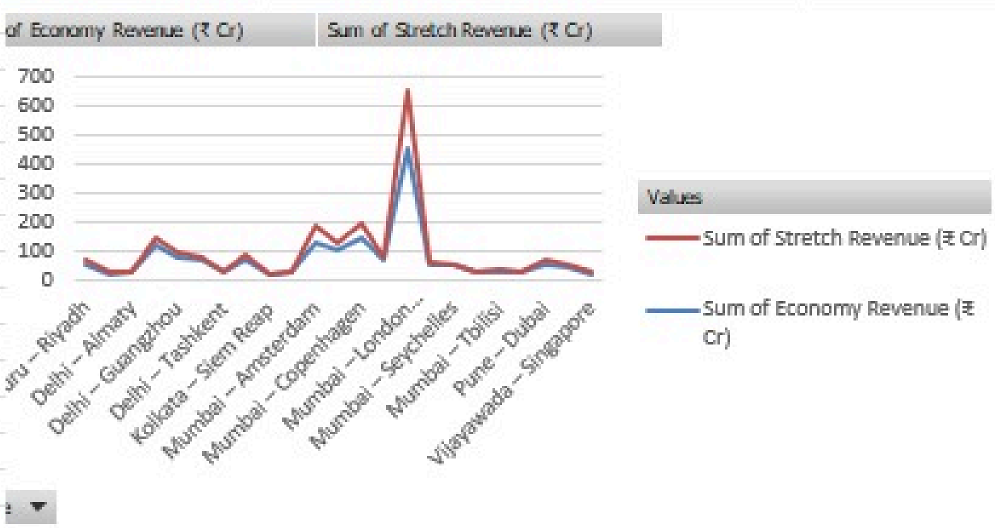


- Pie chart displaying the revenue generation of routes

Row Labels	Sum of Economy Revenue (₹ Cr)	Sum of Stretch Revenue (₹ Cr)
Bengaluru – Riyadh	55.64	10.35
Bhubaneswar – Abu Dhabi	20.64	3.65
Delhi – Almaty	27.69	3.93
Delhi – Athens	117.23	26.52
Delhi – Guangzhou	80.93	12.41
Delhi – Hanoi	67.38	9.86
Delhi – Tashkent	26.33	3.73
Kolkata – Guangzhou	73.06	11.68
Kolkata – Siem Reap	18.09	2.5
Mumbai – Almaty	24.94	3.65
Mumbai – Amsterdam	131.04	56.16
Mumbai – Athens	104.21	25.19
Mumbai – Copenhagen	142.08	56.16
Mumbai – Denpasar (Bali)	66	10.22
Mumbai – London (Heathro	454.56	192.72
Mumbai – Madinah	49.6	9.13
Mumbai – Seychelles	48.92	5.62
Mumbai – Tashkent	24.36	3.48
Mumbai – Tbilisi	31.36	5.62
Pune – Bangkok	26.58	3.93
Pune – Dubai	55.12	10.95
Thiruvananthapuram – Mal	44.03	4.67
Vijayawada – Singapore	22.16	4.03
Grand Total	**1711.95**	**476.16**



- Pivot charts and tables indicating revenue generated from economy/stretch options in the above routes

RouteA1:G30	Aircraft	Flights/Year	Annual Revenue (₹ Cr)	Economy Revenue (₹ Cr)	Stretch Revenue (₹ Cr)	Annual Cost (₹ Cr)	Profit/LOSS (₹ Cr)	Margin
Mumbai – Tbilisi	1	156	41.02	31.36	5.62	79.49	-38.47	-93.8%
Mumbai – Copenhagen	2	156	222.08	142.08	56.16	265.67	-43.59	-19.6%
Mumbai – Denpasar (Bali)	1	365	89.96	66	10.22	190.74	-100.78	-112.0%
Kolkata – Guangzhou	3	365	96.75	73.06	11.68	136.51	-39.76	-41.1%
Mumbai – London (Heathrow)	2	365	697.59	454.56	192.72	674.88	22.71	3.3%
Thiruvananthapuram – Malé	1	365	57	44.03	4.67	79.49	-22.49	-39.4%
Pune – Bangkok	1	156	35.13	26.58	3.93	63.57	-28.44	-81.0%
Pune – Dubai	1	365	77.97	55.12	10.95	131.29	-53.32	-68.4%
Delhi – Guangzhou	1	365	106.82	80.93	12.41	167.11	-60.29	-56.4%
Kolkata – Siem Reap	3	156	23.96	18.09	2.5	49.06	-25.1	-104.8%
Mumbai – Madinah	1	365	68.42	49.6	9.13	152.53	-84.11	-123.0%
Vijayawada – Singapore	1	156	29.44	22.16	4.03	56.34	-26.9	-91.5%
Bengaluru – Riyadh	1	365	74.32	55.64	10.35	135.01	-60.69	-81.6%
Delhi – Hanoi	1	365	88.8	67.38	9.86	172.67	-83.87	-94.5%
Mumbai – Athens	4	156	143.63	104.21	25.19	125.87	17.76	12.4%
Delhi – Athens	4	156	157.98	117.23	26.52	129.28	28.7	18.2%
Bhubaneswar – Abu Dhabi	1	156	28.93	20.64	3.65	57.3	-28.37	-98.0%
Delhi – Almaty	1	156	36.61	27.69	3.93	65.94	-29.33	-80.1%
Mumbai – Almaty	1	156	33.21	24.94	3.65	70.68	-37.46	-112.8%
Delhi – Tashkent	1	156	34.88	26.33	3.73	62.85	-27.97	-80.2%
Mumbai – Tashkent	1	156	32.44	24.36	3.48	68.94	-36.5	-112.5%
Mumbai – Amsterdam	2	156	187.2	131.04	56.16	271.36	-84.16	-45.0%
Mumbai – Seychelles	1	208	61.22	48.92	5.62	97.36	-36.14	-59.1%

Aircraft name	No. assigned
A320 (12J/168Y)	1
787-9 (Leased)	2
A320neo (12J/168Y)	3
A321XLR (20J/190Y)	4

	Aircraft	Flights/Year	Annual Revenu	Economy	Stretch R	Annual Cost (₹ Cr)
Aircraft	1					
Flights/Year	-0.207109866	1				
Annual Revenue (₹ Cr)	0.288887147	0.262067196	1			
Economy Revenue (₹ Cr)	0.306896365	0.275038541	0.999048054	1		
Stretch Revenue (₹ Cr)	0.248768229	0.186978534	0.991120592	0.986306	1	
Annual Cost (₹ Cr)	0.167227998	0.375006374	0.972557719	0.971515	0.965553	1

- Correlation matrix

Routes	Aircraft	Flights/Year	Annual Revenue (₹ C	Economy Revenue (₹ Cr)	Stretch Revenue (₹ C	Annual Cost (₹ Cr	Profit/LOSS (₹ C	Marg
Mumbai – Tbilisi	A320 (12J/168Y)	156	41.02	31.36	5.62	79.49	-38.47	-93.8%
Mumbai – Copenhagen	787-9 (Leased)	156	222.08	142.08	56.16	265.67	-43.59	-19.6%
Mumbai – Denpasar (Bali)	A320 (12J/168Y)	365	89.96	66	10.22	190.74	-100.78	-112.0%
Kolkata – Guangzhou	A320neo (12J/168Y)	365	96.75	73.06	11.68	136.51	-39.76	-41.1%
Mumbai – London (Heathrow)	787-9 (Leased)	365	697.59	454.56	192.72	674.88	22.71	3.3%
Thiruvananthapuram – Malé	A320 (12J/168Y)	365	57	44.03	4.67	79.49	-22.49	-39.4%
Pune – Bangkok	A320 (12J/168Y)	156	35.13	26.58	3.93	63.57	-28.44	-81.0%
Pune – Dubai	A320 (12J/168Y)	365	77.97	55.12	10.95	131.29	-53.32	-68.4%
Delhi – Guangzhou	A320 (12J/168Y)	365	106.82	80.93	12.41	167.11	-60.29	-56.4%
Kolkata – Siem Reap	A320neo (12J/168Y)	156	23.96	18.09	2.5	49.06	-25.1	-104.8%
Mumbai – Madinah	A320 (12J/168Y)	365	68.42	49.6	9.13	152.53	-84.11	-123.0%
Vijayawada – Singapore	A320 (12J/168Y)	156	29.44	22.16	4.03	56.34	-26.9	-91.5%
Bengaluru – Riyadh	A320 (12J/168Y)	365	74.32	55.64	10.35	135.01	-60.69	-81.6%
Delhi – Hanoi	A320 (12J/168Y)	365	88.8	67.38	9.86	172.67	-83.87	-94.5%
Mumbai – Athens	A321XLR (20J/190Y)	156	143.63	104.21	25.19	125.87	17.76	12.4%
Delhi – Athens	A321XLR (20J/190Y)	156	157.98	117.23	26.52	129.28	28.7	18.2%
Bhubaneswar – Abu Dhabi	A320 (12J/168Y)	156	28.93	20.64	3.65	57.3	-28.37	-98.0%
Delhi – Almaty	A320 (12J/168Y)	156	36.61	27.69	3.93	65.94	-29.33	-80.1%
Mumbai – Almaty	A320 (12J/168Y)	156	33.21	24.94	3.65	70.68	-37.46	-112.8%
Delhi – Tashkent	A320 (12J/168Y)	156	34.88	26.33	3.73	62.85	-27.97	-80.2%
Mumbai – Tashkent	A320 (12J/168Y)	156	32.44	24.36	3.48	68.94	-36.5	-112.5%
Mumbai – Amsterdam	787-9 (Leased)	156	187.2	131.04	56.16	271.36	-84.16	-45.0%
Mumbai – Seychelles	A320 (12J/168Y)	208	61.22	48.92	5.62	97.36	-36.14	-59.1%

- Conditional formatting

RESULTS AND INTERPRETATION

Across all 23 new routes in the dataset, the total annual revenue amounted to ₹2425.36 Cr. The total annual operating cost was significantly higher at ₹3303.94 Cr. This results in a net combined loss of approximately ₹878.57 Cr.

This finding indicates that, collectively, the new international routes are currently not profitable, primarily due to high operational expenses and limited average revenue per route.

The average annual revenue per route is ₹105.45 Cr, while the average annual cost per route is ₹143.65 Cr, showing a structural gap between revenue and cost across several medium-haul routes.

Route-Level Results: Profit vs Loss

The route-level analysis revealed significant variation in financial outcomes:

➔ *Highly Profitable / Near-Profitable Routes*

Only a few routes showed positive or near-positive financial results:

- Mumbai – London (Heathrow)

  Revenue: ₹697.59 Cr

  Profit: ₹22.71 Cr

  Margin: +3.3%

- Delhi – Athens

  Revenue: ₹157.98 Cr

  Profit: ₹28.70 Cr

  Margin: +18.2%

- Mumbai – Athens (A321XLR)

Revenue: ₹143.63 Cr

Profit: ₹17.76 Cr

Margin: +12.4%

These routes succeed because they:

- Operate long-haul or premium aircraft (787-9 or A321XLR)
-
- Generate strong business-class (Stretch) revenue
-
- Serve high-demand markets with better yields

This highlights that long-haul routes with premium cabin revenue potential perform better financially.

➔ *Routes With Major Losses*

Many medium-haul A320 routes recorded significant losses. Examples include:

- Mumbai – Denpasar (Bali)

  Loss: –₹100.78 Cr

  Margin: –112%

- Delhi – Hanoi

  Loss: –₹83.87 Cr

  Margin: –94%

- Delhi – Guangzhou

  Loss: –₹60.29 Cr

- Mumbai – Madinah

  Loss: –₹84.11 Cr

- Pune – Dubai

  Loss: –₹53.32 Cr

These losses are driven by:

High annual operating costs for A320 aircraft on medium-haul flights

Lower fares and weaker premium revenue

Lower business demand on these sectors

The findings suggest that certain medium-haul routes may need pricing adjustments, demand stimulation, or aircraft reassignment.

Aircraft Type Performance

The aircraft-wise cost analysis shows that:

A320 (12J/168Y) contributes the highest total annual operating cost (₹1651.31 Cr) because it is used on the largest number of routes.

787-9 (Leased) has high operating cost (₹1211.91 Cr) but also generates strong long-haul revenue.

A320neo has the lowest cost footprint (₹185.57 Cr) but also supports lower-revenue routes.

A321XLR strikes a good balance with moderate cost (₹255.15 Cr) and healthy premium revenue.

Overall, wide-body and long-range aircraft (787-9, A321XLR) outperform narrow-body aircraft on profitability due to higher fare potential and stronger international demand.

Revenue Composition: Economy vs Stretch

The comparison of economy vs stretch-class revenue produced two important insights:

Economy revenue dominates with ₹1711.95 Cr, accounting for ~78% of total annual revenue.

Stretch (Business Class) revenue contributes ₹476.16 Cr (~22%), but is critical for profitability on long-haul routes.

Routes using 787-9 and A321XLR show a much stronger contribution from business-class seats. This reinforces the need for premium cabins on longer international routes.

Revenue vs Cost Pattern

The combined line chart comparing annual revenue and cost across routes clearly shows:

Only 3 routes have revenue lines exceeding cost lines.

The majority of A320-operated routes have overlapping or cost-dominant lines, indicating losses.

High spikes in cost are associated with long-haul aircraft and high-frequency routes, but these routes also show proportionately higher revenue.

This visual comparison supports the conclusion that premium, long-haul routes are financially sustainable, while many medium-haul A320 routes require strategic review.

Regression and Correlation Insights

Correlation analysis revealed that:

Economy revenue has the strongest correlation with total revenue (0.99) → the primary driver of route income.

Stretch revenue shows moderate correlation (0.86) → important but secondary driver.

Flights per year show weak correlation (0.26) → frequency alone does not create profitability.

Aircraft type shows minimal correlation with revenue → meaning demand, pricing, and route characteristics matter more.

This indicates that simply increasing flights or using specific aircraft types does not guarantee profitability; rather, revenue quality and route demand play a larger role.

CONCLUSION

Our Excel-based analysis of IndiGo's FY2022–25 financial data shows that the airline's international expansion is the key driver of its near-term sales growth. Pivot tables and regression models indicate a clear upward trend in revenue tied to new long-haul routes and larger aircraft. This matches company data: Indigo has nearly doubled its overseas network (now 40 international destinations, up 7 from last year), and management confirms that "nearly all" capacity growth is being directed to international markets. For example, the airline has placed large orders for 70 Airbus A321XLRs and leased Boeing 787s to open new Europe and Asia routes. Our six-month forecast predicts rising monthly revenues as these routes ramp up, consistent with Indigo's expectation of high-teens ASK growth this winter. In short, the analysis confirms that Indigo's new A320/A321XLR/787-9 services should substantially boost sales in the coming period, even though the profitability is low at this moment.

However, it also shows that profit margins will remain under pressure initially: in FY2024–25 revenue grew 17% year-on-year but fuel, lease and maintenance costs rose faster , narrowing net margins. Thus, while sales are expected to climb, the forecast accounts for higher operating expenses in the short term.

In summary, our project finds that Indigo's international route push plays a critical role in sales forecasts. The projected sales growth over the next six months aligns with industry reports and company strategy emphasizing long-haul expansion. These insights underscore that the new A321XLR and 787 services should drive revenue growth, even as the airline manages higher costs. Overall, Indigo's internationalization strategy is likely to improve profitability over time by unlocking new passenger markets, validating the forecasts of rising sales with continued expansion.

FUTURE SCOPE

- **Incorporate finer-grained data:** Expand the model to use monthly or quarterly operational data (with passenger volumes, load factors) so forecasts can adjust more quickly to actual performance of new routes.

- **Add external variables:** Include economic and market factors such as fuel prices, foreign exchange rates, and GDP or tourism trends to refine the regression model's accuracy.

- **Use advanced forecasting methods:** Compare the Excel regression results with time-series or machine learning models (ARIMA, random forests, etc.) to potentially capture nonlinear trends and interactions.

- **Perform scenario analysis:** Test the impact of different assumptions (e.g. high/low fuel costs, demand shocks, new competition) to quantify risk to sales and profit projections.

- **Update with real outcomes:** As the new international flights begin operations, incorporate actual revenue and cost figures to validate and recalibrate the forecast model. Tracking performance against predictions will improve future accuracy.

- **Broaden market analysis:** Consider competitive actions (e.g. other carriers' route plans, pricing) and regulatory factors, and extend the horizon beyond six months to capture longer-term seasonality and growth patterns.

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